

**FITNESSWISE**

Ramji Srinivasan

FOR MORE
STORIES

The science of injury versus load management

With advancements in sports medicine and data-driven training methods, India must **develop its own sports science ecosystem** to create a tailored process for data collection and load management.

In 1989, Stephen Covey published his highly acclaimed book, *The 7 Habits of Highly Effective People*. This book, which has sold over 25 million copies, explores how individuals contribute to successful organisations by fostering interdependence among staff.

With the rapid growth of sports medicine teams, sport has become an 'arms race', with team owners often believing that the secret to developing a winning franchise lies in the quality of the backroom staff.

This topic will persist until all related doubts are progressively resolved, taking diverse variables into consideration at various time frames. A practical approach is key.

It's a battle of data versus application. Over the last two decades, research on the relationship between training load and injuries has grown exponentially.

In India, this trend has become prominent in the last four to six years, showing where the country stands in the practical application and data collection in sports science. As we gather more data, our understanding of training modules has improved. However, there is still a long way to go in collecting data relevant to Indian athletes. No data is wasted if it can be used to achieve peak performance with the goal of winning.

The entire thought process has gone through a paradigm shift in the last two decades. Earlier, injuries were directly related to training intensity, load, incorrect methods, and exercise choices. Now, injuries can be attributed to sudden spikes in training load when an athlete is accustomed to a certain load.

In this context, it appears that overuse

injuries may arise from athletes being underprepared for the load they are about to perform. A question of interest to both strength and conditioning (S&C) coaches/physios and researchers is why some athletes sustain injuries at low training loads, while others can tolerate much greater loads. What differences change the perspective in training protocols?

Incorporating a higher chronic training load, well-developed aerobic fitness, and good lower body strength appear to moderate the training-injury relationship and provide a shielding effect against spikes in load.

The training-performance conundrum is complex and dynamic, with multiple inputs to injury and performance at any given time. The challenge faced by sports science researchers is obtaining large enough longitudinal data sets to capture the time-varying nature of physiological and musculoskeletal capacities and training-load data to sufficiently inform injury-prevention efforts.

The training to performance enigma can be solved through

